

* Some reminders:

- Relating QM to the problem:

$\hat{H}$: Hamiltonian

Harmonic Oscillator: $\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m \omega^2 \hat{x}^2$

- eigen states

$$\hat{H}|n\rangle = E_n |n\rangle$$

Since

$$Z_1 = \sum_n e^{-\beta E_n}$$

note

$$e^{-\beta \hat{H}} |n\rangle = e^{-\beta E_n} |n\rangle$$

Since $\langle n|n\rangle = 1$

$$\begin{aligned} \langle n| e^{-\beta \hat{H}} |n\rangle &= \langle n| (e^{-\beta E_n} |n\rangle) \\ &= e^{-\beta E_n} \langle n|n\rangle = e^{-\beta E_n} \end{aligned}$$

Thus,

$$Z_1 = \sum_n \langle n| e^{-\beta \hat{H}} |n\rangle$$

- Can also write this as

$$Z_1 = \text{tr}(e^{-\beta \hat{H}})$$

- * For the Harmonic Oscillator

$$E_n = \left(\frac{1}{2} + n\right) \hbar \omega$$

$$n = 0, 1, 2, \dots$$

- Now, important note: here we have

$$Z_1 = \sum_n e^{-\beta E_n} = \sum_n e^{-\beta \left(\frac{1}{2} + n\right) \hbar \omega}$$

$$= \sum_n e^{-\beta \hbar \omega / 2} \cdot e^{-\beta n \hbar \omega}$$

$$= e^{-\beta \hbar \omega / 2} \sum_{n=0}^{\infty} \left(e^{-\beta \hbar \omega} \right)^n$$

- For $0 < |x| < 1$: $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$

proof:

$$\text{let } S_r = 1 + x + \dots + x^r = \sum_{n=0}^r x^n$$

$$\text{Define: } S_\infty \equiv \lim_{r \rightarrow \infty} S_r = \sum_{n=0}^{\infty} x^n$$

then

$$S_{r+1} = 1 + x + \dots + x^r + x^{r+1}$$

$$x \cdot S_r = x + x^2 + \dots + x^r + x^{r+1}$$

$$\rightarrow S_{r+1} - x S_r = 1$$

Take the limit $r \rightarrow \infty$

$$\lim_{r \rightarrow \infty} S_{r+1} = S_\infty \quad \& \quad \lim_{r \rightarrow \infty} S_r = S_\infty$$

So

$$\lim_{r \rightarrow \infty} (S_{r+1} - x S_r) = S_\infty - x S_\infty = S_\infty (1 - x)$$

but this equals 1, thus

$$S_\infty (1 - x) = 1$$

$$S_\infty = \sum_{n=0}^{\infty} x^n = \frac{1}{1 - x}$$

Since $\beta, \hbar, \omega > 0$ then $e^{-\beta \hbar \omega} < 1$

then

$$\sum_{n=0}^{\infty} (e^{-\beta \hbar \omega})^n = \frac{1}{1 - e^{-\beta \hbar \omega}}$$

$$Z_1 = \frac{e^{-\beta \hbar \omega / 2}}{(1 - e^{-\beta \hbar \omega})}$$

* A note on this term: $e^{-\beta \hbar \omega / 2}$

For $n=0$

$$E_0 = (\frac{1}{2} \hbar \omega) \hbar \omega = \frac{1}{2} \hbar \omega$$

this is the zero point energy. Its often easier to measure energies & other quantities Relative to this state. As such, we simply only use

$$\tilde{E}_n = n \hbar \omega = \hbar \omega, 2 \hbar \omega, \dots$$

instead. So, the idea is that $E_0 = \frac{1}{2}\hbar\omega$ is our "zero"

$$\begin{aligned} E_2 & \sim (\frac{1}{2} + 2)\hbar\omega \longrightarrow 2\hbar\omega \hat{E}_2 \\ E_1 & \sim (\frac{1}{2} + 1)\hbar\omega \longrightarrow 1\hbar\omega = \hat{E}_1 \\ E_0 & \sim \frac{1}{2}\hbar\omega \longrightarrow "0" \end{aligned}$$

As such, we are only interested in the relative partition function

$$Z_1 = e^{-\beta\hbar\omega/2} \tilde{Z}_1$$

$$\boxed{\tilde{Z}_1 = \frac{1}{1 - e^{-\beta\hbar\omega}}} \quad \left(\begin{array}{c} \text{Single Harmonic} \\ \text{Oscillator} \end{array} \right)$$

* This is for a Single harmonic oscillator. *

* Discrete to Continuum:

For multiparticle systems $\tilde{Z} = \sum_n e^{-\beta E_n}$ can be much more complicated to sum. We make the approximation

$$\sum_n \approx \int d^3n = \frac{V}{(2\pi)^3} \int d^3k$$

Spherical coordinates:

$$\begin{aligned} \int d^3k &= \int_0^\infty \int_0^\pi \int_0^{2\pi} |k|^2 \sin(\theta) dk d\theta d\phi \\ &= \left(\int_0^\infty |k|^2 dk \right) \left(\int_0^\pi \sin(\theta) d\theta \right) \left(\int_0^{2\pi} d\phi \right) \\ &= \left(\int_0^\infty |k|^2 dk \right) (2) (2\pi) \\ &= 4\pi \int_0^\infty |k|^2 dk \end{aligned}$$

$$\downarrow \quad \frac{V}{(2\pi)^3} \int d^3k = \frac{4\pi V}{(2\pi)^3} \int_0^\infty k^2 dk = \int_0^\infty \frac{V}{2\pi^2} k^2 dk$$

$$\frac{V}{\pi^2} k^2 dk = D(k) dk \quad (\text{See below})$$

$D(k)$: density of electromagnetic modes

* Putting this together: If we have a blackbody with a large number of particles which emit as a harmonic oscillator:

$$Z_{\text{tot}} = \sum_{n_i} e^{-\beta E_{n_i}}$$

$n_i \leftarrow$ Over all states

example: If we have a two emitter system: E_1 and E_2

$$\tilde{Z}_{\text{tot}} = \sum_{n_1, n_2} e^{-\beta(E_1 + E_2)} \quad |n_1\rangle \quad |n_2\rangle$$

$$= \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} e^{-\beta E_1} \cdot e^{-\beta E_2}$$

$$= \left(\sum_{n_1=0}^{\infty} e^{-\beta E_1} \right) \left(\sum_{n_2=0}^{\infty} e^{-\beta E_2} \right)$$

$$= \tilde{Z}_1 \tilde{Z}_2$$

Thus, if the energies $E_1 = E_1 + E_2$ split then so too does $\tilde{Z}_{\text{tot}}$

$$\ln(\tilde{Z}_{\text{tot}}) = \ln(\tilde{Z}_1) + \ln(\tilde{Z}_2)$$

* Now, for a black body:

$$\tilde{Z}_{\text{tot}} = \tilde{Z}_{1,1} \tilde{Z}_{2,1} \dots$$

$$\ln(\tilde{Z}_{\text{tot}}) = \sum_i \ln(\tilde{Z}_{\omega_i})$$

Replace with integral

$$= \int D(k) \ln(\tilde{Z}_{\omega}) dk$$

$$\ln(\tilde{Z}_{\text{tot}}) = \int \frac{V}{\pi^2} |k|^2 \ln(\tilde{Z}_{\omega}) dk$$

Since

$$F_{\text{tot}} = -k_B T \ln(\tilde{Z}_{\text{tot}})$$

I forgot to mention one thing here
the integral gives a factor

$$\frac{V}{2\pi^2} |k|^2$$

however a photon has two polarizations
and to account for this we multiply this
result by 2:

$$2 \int_0^{\infty} \frac{V}{2\pi^2} |k|^2 dk = \int_0^{\infty} \frac{V}{\pi^2} |k|^2 dk = \int_0^{\infty} D(k) dk$$

Back to problem:

$$F_{\text{tot}} = -k_B T \ln(\tilde{Z}_{\text{tot}}) = -k_B T \int \frac{V}{\pi^2} |k|^2 \ln(\tilde{Z}_{\omega}) dk$$

$$\text{we found } \tilde{Z}_{\omega} = (1 - e^{-\beta \hbar \omega})^{-1}$$

since

$$(1 - e^{-\beta \hbar \omega})^{-1} = \sum_{n=0}^{\infty} e^{-n \beta \hbar \omega}$$

$$\omega = ck, \quad \beta = (k_B T)^{-1}$$

$$\underline{Z_\omega} = \left[1 - \exp\left(\frac{-\hbar ck}{k_B T}\right) \right]^{-1}$$

Plug this in

$$\ln(Z_\omega) = -\ln\left(1 - \exp\left(\frac{-\hbar ck}{k_B T}\right)\right)$$

$$F_{\text{tot}} = \frac{k_B T V}{\pi^2} \int_0^\infty k^2 \ln\left[1 - \exp\left(\frac{-\hbar ck}{k_B T}\right)\right] dk$$

First part of problem is simply
 $F = -k_B T \ln(Z)$

(b) i) Internal (Average) energy

$$Z = \sum_n e^{-\beta E_n}$$

From QM: the probability of a system to be found in $|n\rangle$ is

$$P(n) = e^{-\beta E_n} / Z$$

(internal)

$$\text{Avg. Energy: } \langle \hat{H}_n \rangle = \sum_n E_n P(n) = U_{\text{int}}$$

$$\langle \hat{H}_n \rangle = \sum_n E_n e^{-\beta E_n} / Z$$

$$= \frac{1}{Z} \sum_n E_n e^{-\beta E_n} = \frac{1}{Z} \sum_n -\frac{\partial}{\partial \beta} (e^{-\beta E_n})$$

$$= -\frac{1}{Z} \frac{\partial}{\partial \beta} \left(\sum_n e^{-\beta E_n} \right) = -\frac{1}{Z} \frac{\partial Z}{\partial \beta}$$

note

$$\frac{\partial}{\partial \beta} \ln(z) = \frac{1}{z} \frac{\partial z}{\partial \beta}$$

thus

$$U_{\text{int}} = \langle \hat{A}_n \rangle = - \frac{\partial \ln(\tilde{z})}{\partial \beta}$$

(ii) Since $F = -k_B T \ln(z) = -\beta^{-1} \ln(z)$
 $\ln(z) = \beta F$
plugging this into the above:

$$U_{\text{int}} = - \frac{\partial}{\partial \beta} (\beta F)$$

entropy:

$$\begin{aligned} S &= k_B (\ln(z) + \beta U_{\text{int}}) \\ &= k_B (-\beta F + \beta U_{\text{int}}) \\ &= \beta k_B (-F + U_{\text{int}}) \\ &= \frac{k_B}{k_B T} (F + U_{\text{int}}) \end{aligned}$$

thus

$$S = \frac{1}{T} (-F + U_{\text{int}})$$

(iii) For the expressions just use the above.
To check your answer for the entropy S , you can use

$$S = - \frac{\partial F}{\partial T}$$

$$U_{int} = - \frac{\partial}{\partial \beta} (\beta F) = - \frac{\partial}{\partial \beta} \left(\frac{V}{\pi^2} \int dk k^2 \ln(1 - e^{-\beta \hbar c k}) \right)$$

$$= - \frac{V}{\pi^2} \int dk k^2 \left(\frac{1}{1 - e^{-\beta \hbar c k}} \right) \cdot \hbar c k e^{-\beta \hbar c k}$$

$$U_{int} = - \frac{V \hbar c}{\pi^2} \int_0^\infty dk \frac{k^3}{1 - e^{-\beta \hbar c k}} \cdot e^{-\beta \hbar c k}$$

$$S = \frac{1}{T} \left(- \frac{k_B T V}{\pi^2} \int dk k^2 \ln(1 - e^{-\beta \hbar c k}) - \frac{V \hbar c}{\pi^2} \int_0^\infty dk \frac{k^3 e^{-\beta \hbar c k}}{1 - e^{-\beta \hbar c k}} \right)$$

